FIU Knight Foundation School of Computing and Information Sciences
FLORIDA INTERNATIONAL UNIVERSITY

# Dayform - AI-Powered Fitness & Wellness Tracker

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**COMPUTER SCIENCE**
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## Problem

• Many students and busy adults struggle to track workouts, nutrition, and wellness goals consistently.

• Using separate apps or notes makes it harder to understand overall progress.

• Users need a simple, centralized way to monitor their habits and stay organized.

• Many tracking tools focus on recording individual activities rather than showing how workouts, nutrition, and goals relate to overall progress.

• Without clear progress summaries and personalized feedback, users may struggle to identify patterns and maintain consistent habits.

## Stakeholders & Requirements

### Stakeholders

- Students and busy adults looking to manage their fitness and wellness.
- Users who want to track their progress in one place.

### Functional Requirements

- Secure accounts and personal profiles.
- Workout, nutrition, and wellness goal tracking.
- Progress dashboards and AI-powered wellness insights.

### Non-Functional Requirements

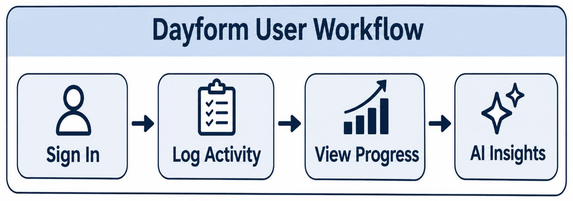
- Secure, user-specific data storage.
- Responsive design for desktop and mobile.
- Reliable performance and maintainable code.

## Our Solution

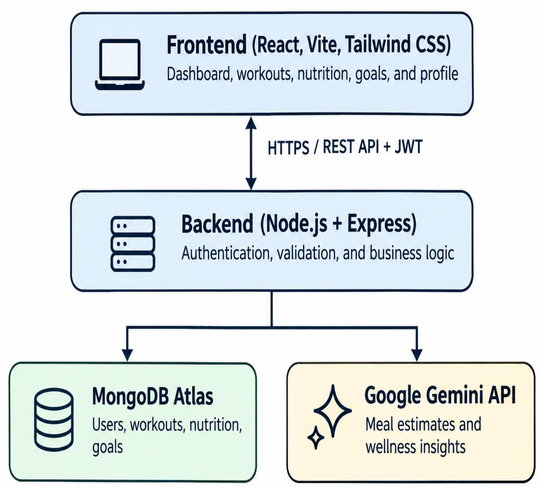
Dayform is a web-based fitness and wellness tracker that brings workouts, nutrition, and personal goals into one application.

### Key Design Decisions

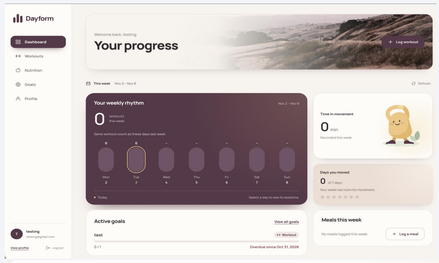
- **Centralized Dashboard:** Displays activity, nutrition, and goal progress in one place.
- **Secure User Accounts:** Keeps each user's information separate through authenticated access.
- **AI Integration:** Uses Google Gemini to provide personalized wellness insights and meal estimates based on user-provided information.



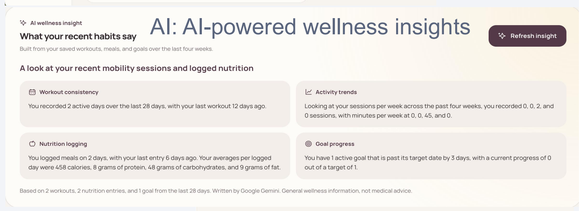

## Architecture



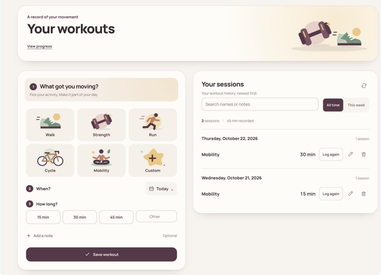

## Demo & Screenshots




Fitness and wellness dashboard




AI: AI-powered wellness insights




Workout logging and history

## Results & Evaluation

| 58/58 | 15/15 | 73/73 |
|---|---|---|
| Backend Tests | Frontend Tests | Integration Tests |

146 tests passed in the latest verified runs

### Implemented Features

- Secure registration, login, and user profiles.
- Workout logging, nutrition tracking, and goal management.
- Dashboard summaries and AI-powered wellness insights.
- Deployed frontend and backend on Vercel.

### Testing Results

- 146 tests passed across the latest verified test runs.
- Production build completed successfully on November 2, 2026.

## Engineering Process (SO6)

### Development Process

• Followed Agile Scrum across five planned sprints.

• Managed user stories, tasks, and acceptance criteria in the official Capstone website.

• Used GitHub for version control and code reviews.

### Testing & Evaluation

• Developed backend, frontend, and integration tests.

• Used automated testing and manual application walkthroughs.

• Reviewed results and addressed issues throughout development.

## What's Next

### Future Improvements

- Integrate wearable devices to automatically track workouts and daily activity.
- Expand AI-powered recommendations using long-term fitness and nutrition trends.
- Add more detailed progress reports and personalized goal tracking.
- Continue improving accessibility, performance, and the mobile experience.

## People & Acknowledgments

**Team Leader:** Alejandro Perez-Rivero

**Team Members:**
- Alejandro Perez-Rivero
- Ahanaf Akif
- Allen Cruz

**Product Owner:** Alejandro Perez-Rivero

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